Exercise-policy loss under computational and financial constraints

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Abstract

An approximate exercise policy must be assessed against the best expected reward attainable under the same financial conventions and exercise rights, although that benchmark is generally unknown. I develop a consistent evaluation design for this problem in a finite-calendar stopping model with deterministic funding and discrete cash events. Conditional martingale arguments separate policy loss from dual slack, while calendar and policy comparisons distinguish changed opportunities from a feasible terminal-policy gain. Repeated-training inference separates variation between fitted policies from evaluation noise, and complete-workload timing assesses the expenditure forecasts used for allocation. In thirteen related development specifications, analytically centred value functions reduce estimated gaps by 91.42 to 95.23 per cent without changing the exercise policies. A prespecified computational follow-up shifts evaluation effort towards more independent training pairs; its selected estimator-variance estimates are 16.7 to 34.1 per cent of their reference estimates. Historical cost forecasts nevertheless fail to transfer. Local calibration reduces mean absolute relative CPU forecast error from 29.065 to 2.841 per cent, but retains an overrun and incurs additional cost. These are conditional simulation findings, not evidence of market profitability or reliable budget protection. The attempted independent market-validation sample yields no admissible observations. I therefore delimit the completed computational study from the unresolved empirical validation.

Keywords: optimal stopping; American options; martingale duality; Monte Carlo; discrete dividends; computational allocation.

1 Introduction

For an early-exercise put, an approximate rule chooses whether to exercise at each admissible right using the information then available. The practical question is how much expected discounted reward that rule gives up, and how reliably that amount can be assessed within a computational budget. The optimal rule is generally unknown. A smaller measured gap may reflect a tighter evaluation bound rather than better exercise decisions; altering funding, cash entitlements or exercise rights may instead change the comparison problem. Fitting more policies, evaluating them more precisely and constructing tighter bounds also compete for computation.

The contribution of this paper is to organise these comparisons within one consistent evaluation design for a funded cash-event stopping model. I separate policy loss from dual slack while holding the exercise instructions fixed. I distinguish enlarged exercise opportunities from a feasible terminal-policy adjustment and from the additional effect of retraining. Conditional inference states which model inputs, fitted functions and sampling choices each comparison retains. Repeated-training evaluation separates variation between fitted policies from evaluation noise, and complete-workload timing tests the expenditure forecasts used to allocate computation. The design connects financial rewards, mathematical bounds, inferential precision and measured costs so that a change in one can be interpreted without attributing it to another.

Within this design, I ask whether a tighter loss assessment reflects a change in policy reward, how computation should be divided between independent training repetitions and evaluation

paths, and whether measured costs predict expenditure for the resulting complete workloads. A finely resolved PDE price supplies a numerical comparison, but is not treated as a known optimum without a justified error enclosure.

The methodological foundations are established. Longstaff and Schwartz (2001) construct exercise policies through backward continuation regression. Haugh and Kogan (2004) develop martingale upper bounds and the use of approximate value processes. The lognormal hinge expectation used below follows the analytical pricing foundation of Black and Scholes (1973). I apply these foundations to an explicitly funded cash-event model and organise the resulting policy comparisons. I do not claim the dual inequality, conditional Jensen argument or finite-family inference as new general theorems.

Pricing intervals and policy assessment are already central to Andersen and Broadie (2004). Avoiding nested simulation is also established: Belomestny et al. (2009) construct martingale-preserving estimators, and Cline (2026) studies regression-based duals with explicit conditional restrictions. Consequently, analytical centring and lower computational cost do not by themselves identify a new contribution. I use the funded cash-event setting to separate changes in policy reward, evaluation slack and exercise opportunities. A claim about their economic importance requires controlled evidence on each margin.

Appendix A applies predictable compensation to penalties with conditional drift. It gives a loss allowance and a paired-comparison correction, together with a finite analytical example. Appendix B specifies the information available to a learned policy and derives the variance of a repeated-training comparison when earlier evidence determines its sample sizes. I present these as consequences of established theory. The literature comparison does not establish priority for the propositions or their proofs.

The evidence concerns the controlled comparisons just described. I retain both the conditional precision gains and the failed expenditure forecasts, then assess local timing calibration on complete, fixed workloads. Appendix C connects uncertainty about the variance components to the implemented allocation rule. The development specifications share a narrow financial setting, and a separate attempt at independent market validation supplies no admissible primary observations. The completed study therefore supports conditional computational conclusions only.

Section 2 defines the financial state, cash transfers and admissible exercise rights. Sections 3 and 4 distinguish loss measurement, changed opportunities and policy gains. Sections 5 and 6 specify conditional inference and computational allocation. Section 7 reports the numerical comparisons, and Section 8 assesses their financial interpretation. The notation guide below collects the principal symbols; Figures 1 and 2 show the event order and evaluation sequence.

2 The funded finite-calendar stopping problem

2.1 State, payments and information

Consider a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{Q})$ carrying a Brownian motion. Let $D_0(u) > 0$ denote a deterministic discount factor for a payment at date u. If a quoted share price S_t settles at $u(t)$, define

$$Y_t = f_t S_t, \qquad f_t = D_0(u(t)). \tag{1}$$

The variable Y represents discounted economic ownership under the maintained settlement convention. This transformation requires the share and exercise transfer to represent matching economic entitlements. It does not establish a financing strategy or identify the correct funding curve from market prices alone.

Between prescribed cash events, I assume

$$dY_t = \sigma Y_t \, dW_t^{\mathbb{Q}}, \qquad \sigma \geq 0. \tag{2}$$

Core notation

Symbols	Meaning
$\mathbb{Q}, \mathcal{F}_j, \mathcal{H}_k$	Maintained pricing measure; information at exercise right j and propagation node k, respectively.
$S_t, Y_t, D_0(u)$	Quoted share price, discounted ownership state and deterministic payment discount factor.
$\mathcal{E}, e_j = (t_j, p_j)$	Admissible rights, ordered by physical date and cash-event phase.
f_t, f_j, b	Share-settlement, exercise-settlement and premium-payment factors, respectively.
c, C	Proposed discounted cash cap and detached entitlement at a cash event.
H_j, π, τ_π	Discounted put reward, exercise policy and its selected right.
V^*, V_π, ℓ_π	Optimal and policy values in discounted currency; only ℓ_π divides their difference by the premium factor b.
v_k, M_j	Retained values on the propagation grid and an integrable martingale restricted to exercise rights.
U, L_π, G_π	Upper payoff, martingale-adjusted policy payoff and their non-negative pathwise gap.
$D, \mathcal{P}$	Data fitting a policy pair and preceding evidence fixing fresh sample counts.
$m, n, \widehat{\mu}$	In the training comparison, independent fitted pairs, paths per pair and mean-gain estimator.
$\mathcal{V}_{\mathrm{tr}}, \mathcal{V}_{\mathrm{ev}}$	Variation between conditional pair means and mean conditional evaluation variance.

At an event with proposed discounted cash $c \geq 0$, the detached entitlement and remaining state are

$$C = \min(Y_{t-}, c), \qquad Y_t = (Y_{t-} - c)^+, \qquad Y_t + C = Y_{t-}. \tag{3}$$

The cap preserves non-negativity and cash accounting. This bounded cash specification is related to the stock-dependent dividend framework discussed by Vellekoop and Nieuwenhuis (2006); the settlement and ownership equations here specify the particular maintained model. The ex-entitlement state is not a martingale across the cash detachment; between events it is. Conditional transition calculations must respect this distinction. Proposed cash, volatility and funding are model inputs, not identified expectations merely because they permit a price fit.

For a put with strike $K > 0$, let $\mathcal{E} = \{e_1, \ldots, e_J\}$ be a finite ordered set of admissible exercise rights, where $e_j = (t_j, p_j)$ records the physical time and prescribed cash-event phase, with $0 < t_1 \leq \cdots \leq t_J$. Rights are ordered by time, with a pre-event right preceding a post-event right when their times coincide. A pre-event phase is used only at a prescribed cash event. Valuation is at $t_0 = 0$.

Write Y_j for the ownership state at right e_j and $\mathcal{F}_j$ for the information available there. At a cash event, these states are Y_{t_j-} before detachment and Y_{t_j} afterwards. Use $\mathcal{F}_j = \mathcal{F}_{t_j-}$ for a pre-event right and $\mathcal{F}_j = \mathcal{F}_{t_j}$ otherwise, with valuation information $\mathcal{F}_0$ and $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \cdots \subseteq \mathcal{F}_J$. Let $f_j = D_0(u_j) > 0$ denote the factor for the exercise settlement specified at that right. Define

$$H_j = (K f_j - Y_j)^+. \tag{4}$$

For quantities indexed by the rights, use $X_{e_j} = X_j$. Thus H_τ and M_τ refer to the selected right, including its phase, rather than to its physical date alone. Admissibility means that $\{\tau_\pi \preceq e_j\} \in \mathcal{F}_j$, where $\preceq$ denotes the stated order of rights. The exercise-settlement factor can differ from the premium-payment factor $b > 0$. Values below are in discounted currency unless divided by b. The definition of H_j presumes that the specified exercise settlement transfers the intended ownership state and entitlements. A change in deliverable or exercise funding can require a different reward.

Assumption 1 (Admissibility)**.** *The decision set is finite and fixed before evaluation. Rewards are adapted and integrable. Policies are stopping times taking values in $\mathcal{E}$. Evaluation martingales start at zero and are integrable at all rights. Whenever sampling variances are reported, the corresponding random variables have finite second moments. Learned coefficients and approximate value functions are fixed before independent evaluation paths are drawn.*

With deterministic initial state and model inputs, the optimal and policy values are

$$V^* = \sup_{\tau \in \mathcal{T}(\mathcal{E})} \mathbb{E}[H_\tau], \qquad V_\pi = \mathbb{E}[H_{\tau_\pi}], \qquad \ell_\pi = \frac{V^* - V_\pi}{b} \geq 0. \tag{5}$$

Cash-event observations needed for propagation do not automatically grant exercise rights. The order of an exercise decision, detachment and settlement is part of the problem definition. A finite exercise calendar represents a Bermudan problem; its numerical refinement alone does not verify the contractual exercise opportunities of an American option.

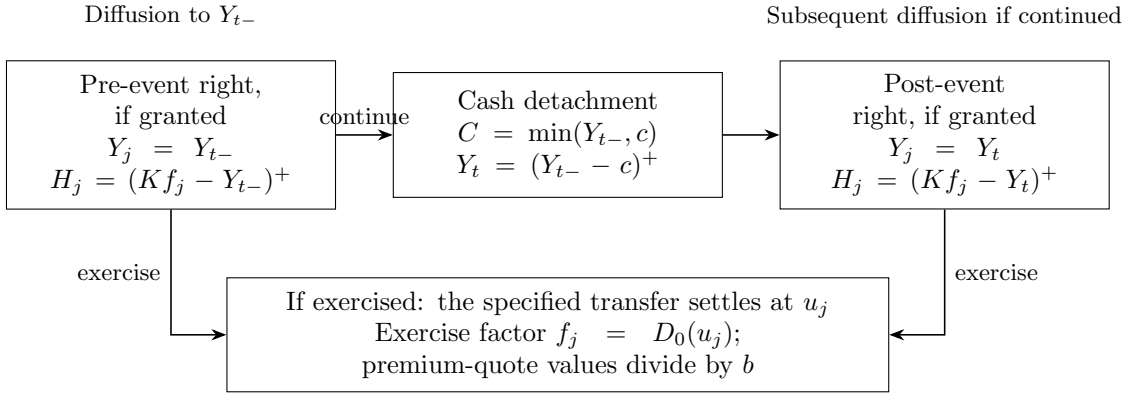


Figure 1: Decision order at a cash event. The calendar grants either, both or neither of the displayed exercise rights. Their factors belong to the selected right. Settlement is shown separately because its date is a payment convention, not an additional simulated right. The cap c already expresses the proposed cash in discounted units; the detached entitlement is not an additional put payoff. Evaluation may retain the full ownership path after policy exercise to calculate the dual maximum.

2.2 Learned exercise instructions

Backward regression estimates continuation values on a training sample and converts them into stopping predicates (Longstaff and Schwartz, 2001). All coefficients, basis choices, positivity safeguards and tie rules form part of the frozen policy. Independent forward evaluation estimates its value without treating its training fit as realised out-of-sample accuracy. Increasing basis dimension need not improve a finite-sample rule.

When a financial scenario θ changes the ownership factor, I preserve the instruction in quoted-price coordinates. If a_j^θ is the scenario share-settlement factor, the quoted state is $Q_j = Y_j^\theta / a_j^\theta$. A source predicate expressed in its original ownership coordinate receives

$$T_j^\theta(Y_j^\theta) = Y_j^\theta \frac{a_j^0}{a_j^\theta}. \tag{6}$$

Its original coefficients and decision conventions are retained, but the resulting stopping time is valued with scenario rewards. This preserves the instruction for an identical history of quoted prices. It does not preserve the distribution of stopping dates when the scenario dynamics change.

3 Loss measurement and dual slack

3.1 A pathwise gap

For an admissible martingale M with $M_0 = 0$, set

$$U(M) = \max_{1 \leq j \leq J} \{H_j - M_j\}, \qquad L_\pi(M) = H_{\tau_\pi} - M_{\tau_\pi}, \qquad G_\pi(M) = U(M) - L_\pi(M). \tag{7}$$

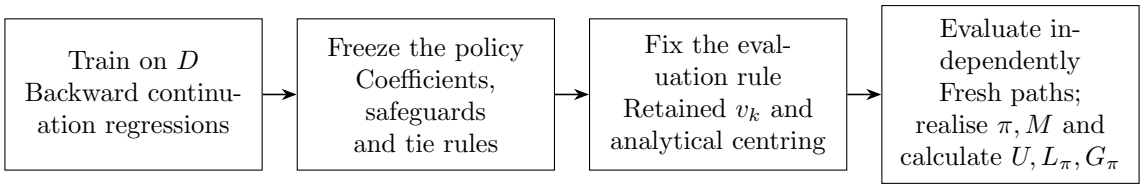


Figure 2: Information fixed before evaluation for the analytically centred value construction. Deterministic value construction can precede training or be shared across policies; the diagram specifies what must be retained before fresh paths are drawn. The realised martingale is computed on those paths, with conditional expectations integrating the retained functions analytically rather than through inner simulations. Training-procedure comparisons repeat complete fitted pairs independently and use common fresh evaluation paths within a pair. Costs refer to the declared workload in Section 6.

Here the maximum ranges over the indexed exercise rights. I use the martingale dual formulation of Haugh and Kogan (2004).

Proposition 1 (What a gap measures)**.** *Under Assumption 1, $G_\pi(M) \geq 0$ pathwise and*

$$\mathbb{E}[G_\pi(M)] = \underbrace{V^* - V_\pi}_{\textit{policy loss}} + \underbrace{\mathbb{E}[U(M)] - V^*}_{\textit{dual slack}}. \tag{8}$$

Both terms on the right are non-negative.

Proof. The maximum defining U includes the policy's stopping right, proving the pathwise inequality. Finite-horizon optional sampling gives $\mathbb{E}[M_\tau] = 0$ for each admissible stopping time. Hence $\mathbb{E}[L_\pi] = V_\pi$. Also $U \geq H_\tau - M_\tau$ for every such τ, so $\mathbb{E}[U] \geq \sup_\tau \mathbb{E}[H_\tau] = V^*$. Subtracting V_π and adding and subtracting V^* gives (8). □

For a fixed policy and financial problem, a change in expected gap caused only by replacing M is a change in dual slack. It is not a change in policy value. A numerical PDE value supplies an additional comparison but is not inserted as an exact V^* without a justified error enclosure. Mesh agreement is evidence about numerical behaviour, not a proof of the financial model.

If two policies use the same upper payoff and martingale, then

$$G_\pi(M) - G_\rho(M) = L_\rho(M) - L_\pi(M), \qquad \mathbb{E}[G_\pi - G_\rho] = V_\rho - V_\pi. \tag{9}$$

The upper payoff cancels pathwise. With the martingale condition satisfied, paired gap differences identify expected policy gains without confounding them with changes in the bound. Appendix A gives the additional compensator term when that condition fails.

3.2 Analytically centred approximate values

The propagation grid includes valuation, exercise rights and the cash-event observations needed between them. Index its ordered time-and-phase nodes by $k = 0, \ldots, N$, write $\overline{Y}_k$ for the state there and $\mathcal{H}_k$ for the available path information, and let $\kappa(j)$ identify exercise right e_j. Thus $Y_j = \overline{Y}_{\kappa(j)}$ and $\mathcal{F}_j = \mathcal{H}_{\kappa(j)}$. Let v_{k+1} be a deterministic approximate value at the next observation, or a function fixed by independent training information. Starting from $\overline{M}_0 = 0$, define

$$\Delta\overline{M}_{k+1} = v_{k+1}(\overline{Y}_{k+1}) - \mathbb{E}[v_{k+1}(\overline{Y}_{k+1}) \mid \mathcal{H}_k], \qquad M_j = \overline{M}_{\kappa(j)}. \tag{10}$$

When the conditional expectation is evaluated for the actual retained function and transition, these increments have conditional mean zero in exact arithmetic. A cash detachment is deterministic given its pre-event state, so its centred increment is zero. Restricting the observation-grid

martingale to the prescribed exercise rights preserves the martingale property; the added observations grant no exercise rights. Across an intervening cash observation, an exercise-to-exercise increment may be a sum of centred terms rather than a single value residual. Approximation quality affects tightness rather than supplying the reason for centring. The construction is the martingale component of an approximate value process described by Haugh and Kogan (2004).

I represent approximate values on the non-negative state domain as

$$v(y) = \alpha + \beta y + \sum_{r=1}^{R} \gamma_r \left(y - k_r\right)^+, \qquad k_r \geq 0. \tag{11}$$

Over a no-cash interval of length h, $X = y\exp(-\sigma^2 h/2 + \sigma\sqrt{h}\,Z)$, with Z standard normal. For $y, k > 0$ and $\sigma^2 h > 0$,

$$\mathbb{E}[(X-k)^+] = y\Phi(d_1) - k\Phi(d_2), \tag{12}$$

$$d_1 = \frac{\log(y/k) + \sigma^2 h/2}{\sigma\sqrt{h}}, \qquad d_2 = d_1 - \sigma\sqrt{h}. \tag{13}$$

Here Φ is the standard-normal distribution function. This is the lognormal option integral (Black and Scholes, 1973). Zero state, strike, variance or time uses its continuous limiting form. Each term in (11) can consequently be integrated over the full support, including the retained tail rule.

For a capped cash jump with size c, composition is explicit:

$$v((x-c)^+) = \alpha + \beta\left(x-c\right)^+ + \sum_{r=1}^{R} \gamma_r \left(x - (c + k_r)\right)^+. \tag{14}$$

The identity uses $k_r \geq 0$ and includes the constant segment for $x \leq c$. Cash-phase ordering must be preserved when multiple propagation intervals are joined. An approximate quadrature expectation or truncated integration range cannot simply be substituted into (10) while retaining an exact martingale claim.

Floating-point evaluation, function compression and approximation of the dynamic programme are distinct numerical errors. Validation should check analytical limiting cases, cash accounting, tail treatment and independent pricing references. A compression tolerance for one function is not an accumulated error bound for the stopping problem.

4 Changed opportunities and changed policies

4.1 Nested calendars with unchanged instructions

Let $\theta \in \Theta$ index a prescribed finite financial family and let d range over a finite ordered endpoint set $\mathcal{D}$ with largest element $d_{\max}$. Let $\mathcal{E}(d)$ be nested exercise calendars. Write $V_\theta^*(d)$ for the optimum and suppose that the original policy always stops on the smallest calendar. Its reward definition and pre-terminal dynamics remain unchanged when d varies within a financial scenario.

Proposition 2 (Maximal-calendar reduction)**.** *If $\mathcal{E}(d_1) \subseteq \mathcal{E}(d_2)$ whenever $d_1 \leq d_2$, and $b_\theta > 0$ is constant across these calendars, then*

$$\max_{\theta\in\Theta,\, d\in\mathcal{D}} \frac{V_\theta^*(d) - V_{\pi,\theta}}{b_\theta} = \max_{\theta\in\Theta} \frac{V_\theta^*(d_{\max}) - V_{\pi,\theta}}{b_\theta}. \tag{15}$$

Proof. Every stopping rule on a smaller calendar is feasible on a larger one, so the optimum is nondecreasing. The original policy value is constant within the scenario because its decisions and rewards remain unchanged. Subtracting this constant and dividing by a positive fixed premium factor preserves monotonicity. Taking the maxima gives (15). □

The statement concerns true expected losses. Sample upper limits need not increase with the calendar because sampling noise and dual slack can change. Evaluating the prespecified maximal calendar supplies a bound for smaller nested calendars; it does not claim to have priced or simulated every intermediate case. Recalibration across endpoints, a changed cash transfer or a policy allowed to use new observations would require a different comparison.

4.2 A feasible terminal adjustment

Suppose the original endpoint is T and the extended endpoint is $T + h$. Retain every original exercise right and add rights only after T. Between the endpoints, the maintained lognormal state Y is an integrable martingale with no cash event. Every added exercise right has the same funded strike $k = Kf_J$ as the original terminal right. The transition depends on the current state rather than on additional past information, so $\mathbb{E}[g(Y_{T+h}) \mid \mathcal{F}_T] = g_h(Y_T)$ for the function defined below. Define

$$g(y) = (k - y)^+, \qquad g_h(y) = \mathbb{E}[g(Y_{T+h}) \mid Y_T = y], \qquad q_h(y) = g_h(y) - g(y). \tag{16}$$

Conditional Jensen gives $q_h(y) \geq 0$. Under the lognormal transition, g_h follows the corresponding analytical put integral. The endpoint condition is stronger than calendar nesting: it provides a tractable terminal replacement and dominance of earlier exercise within the added interval.

Let the compulsory final right of the original policy be $e_J = (T, \text{after})$, so $Y_J = Y_T$ and $\mathcal{F}_J = \mathcal{F}_T$. Let $A = \{\tau_\pi = e_J\} \in \mathcal{F}_J$. Change only the final instruction: if A occurs, wait until $T + h$; otherwise retain the original stopping right. Denote this adjusted policy by $\widetilde{\pi}$.

Proposition 3 (Expected gain from the final instruction)**.** *The adjusted rule is feasible on the expanded calendar and*

$$V_{\widetilde{\pi}} - V_\pi = \mathbb{E}[\mathbf{1}_A q_h(Y_T)] \geq 0. \tag{17}$$

The conditionally integrated gain has variance no greater than the direct realised payoff difference, when the latter has finite variance.

Proof. The decision to wait is made using the $\mathcal{F}_T$-measurable event A, so the rule is a stopping time. Outside A, payoffs agree. On A, the tower property gives the expected difference as $q_h(Y_T)$. Conditional Jensen proves its sign. The integrated gain is the conditional expectation of the realised difference given $\mathcal{F}_T$; the law of total variance proves the variance comparison. □

The realised later payoff can still be smaller. Equation (17) establishes an expected improvement under the maintained model, not a pathwise profit or executable market opportunity.

The implemented primary comparison integrates the extra diffusion before constructing the gap. Conditional Jensen and optional sampling imply that any exercise in the added interval is dominated in conditional expected reward by waiting to $T+h$. The expanded optimum therefore equals the original-clock stopping optimum with terminal reward replaced by $g_h(Y_T)$. Write

$$\begin{aligned} H_j^{(h)} &= \begin{cases} H_j, & j < J, \\ g_h(Y_T), & j = J, \end{cases} \\ U_h^c(M) &= \max_{1 \leq j \leq J} \{H_j^{(h)} - M_j\}, \\ L_{\widetilde{\pi}}^c(M) &= L_\pi(M) + \mathbf{1}_A q_h(Y_T). \end{aligned} \tag{18}$$

Here c denotes conditional integration. Optional sampling on the original clock gives $\mathbb{E}[L_{\widetilde{\pi}}^c] = V_{\widetilde{\pi}}$, and the dual argument for the modified terminal reward gives $\mathbb{E}[U_h^c] \geq V_h^*$. Using U_h^c for both instructions makes their gap difference exactly $\mathbf{1}_A q_h(Y_T)$ on each retained original path. The separate direct diagnostic simulates the later reward and computes its upper payoff on the expanded clock. Its pathwise gaps and sampling variance need not equal those of the integrated

comparison. This distinction explains why the primary comparison isolates the expected policy gain without requiring the additional realised diffusion.

The residual discounted loss satisfies

$$V_h^* - V_{\widetilde{\pi}} = (V_0^* - V_\pi) + (V_h^* - V_0^*) - \mathbb{E}[\mathbf{1}_A q_h(Y_T)]. \tag{19}$$

The policy gain need not be bounded above by the increase in the optimum: a suboptimal original rule may survive where an optimal original rule would already stop. Residual loss for a changing policy need not be monotone in the endpoint.

4.3 Retraining as a separate comparison

Replacing the terminal training response by g_h can also change earlier continuation regressions. To isolate that effect, I use the same training paths, basis, regression safeguards and financial inputs for original and revised fits, and give both policies the same final waiting instruction. I freeze the fits before fresh paired evaluation.

Equation (17) does not imply that finite-sample retraining improves value. Changed stopping decisions alter the responses entering earlier regressions, and estimated continuation can move in either direction. A common upper payoff identifies the incremental expected value through (9). Unfavourable and inconclusive paired outcomes must remain in the comparison.

5 Conditional inference and scenario families

5.1 What the probability statement conditions on

With the financial model, fitted policy and approximate values fixed, simulate independent outer paths. Let $X_i = G_\pi^{(i)}(M)/b$, with sample mean $\overline{X}$ and sample standard deviation s_X. A commonly used approximate upper limit is

$$B_\pi = \overline{X} + t_{1-\alpha,n-1}\frac{s_X}{\sqrt{n}}. \tag{20}$$

Since $\ell_\pi \le \mathbb{E}[X_i]$, a valid upper limit for the expected gap is also an upper limit for policy loss. Student calibration is generally approximate for non-normal gaps; it is not a finite-sample guarantee. Finite variance, independent evaluation and sufficiently informative Monte Carlo samples remain necessary. More outer paths improve precision about the expected gap but do not remove its dual slack.

A conditional interval does not integrate over the uncertainty of policy training, financial calibration, contract conventions or sample selection. Training repetitions must be retained separately if their variation is investigated. Choosing a policy on the same evaluation outcomes used to advertise its bound can invalidate the selected-policy interpretation. Such a selection requires a separate evaluation design or an appropriate simultaneous guarantee.

5.2 Coverage for a prescribed financial family

For $m_\Theta = |\Theta|$, compute a per-scenario limit at the maximal calendar using error allocation α/m_Θ:

$$B_\theta = \overline{X}_\theta + t_{1-\alpha/m_\Theta,n_\theta-1}\frac{s_\theta}{\sqrt{n_\theta}}. \tag{21}$$

If each marginal limit has its stated coverage, the union bound and (15) imply an event of probability at least $1-\alpha$ on which every prescribed scenario and smaller calendar is covered. With approximate marginal calibration, the joint statement inherits that approximation. Independence across scenarios is unnecessary for the union bound.

There is no extra multiplicity factor for calendars already dominated by the maximal endpoint. There is also no automatic extension to all policies, a continuous funding region or scenarios introduced after inspection. A missing required scenario prevents a complete-family conclusion; the family size must not shrink opportunistically. Financial quote compatibility is a different condition and must be reported separately from mathematical completeness of the conditional family.

5.3 Pairing, independent runs and gap interactions

Within a common-path experiment, I estimate policy gains or scenario effects from pathwise differences. For $D_i = X_i - Y_i$,

$$\widehat{\mathrm{Var}}(\overline{D}) = \frac{s_D^2}{n} = \frac{s_X^2 + s_Y^2 - 2s_{XY}}{n}. \tag{22}$$

Discarding covariance would answer a different uncertainty question. Conversely, unrelated evaluation streams are not paired merely because policies or model labels coincide.

For independent mean estimates R and L, possibly with different path counts,

$$\widehat{v} = \frac{s_R^2}{n_R} + \frac{s_L^2}{n_L}, \qquad \nu = \frac{\widehat{v}^2}{(s_R^2/n_R)^2/(n_R-1) + (s_L^2/n_L)^2/(n_L-1)}. \tag{23}$$

The usual Welch interval uses $t_{1-\alpha/2,\nu}\sqrt{\widehat{v}}$. Degenerate zero-variance cases require explicit treatment. A difference between reported confidence endpoints is a descriptive comparison, not itself a confidence interval for a policy-value difference.

A financial/calendar interaction can be written in terms of expected quoted gaps as

$$I_\theta = \{g_\theta(d_{\max}) - g_0(d_{\max})\} - \{g_\theta(0) - g_0(0)\}. \tag{24}$$

When each financial effect is paired internally but the two calendar experiments are independent, add the variances of their paired mean effects. By (8), I_θ combines interactions in true loss and dual slack. It does not identify either component alone and has no causal interpretation. Repeated dependent pointwise comparisons do not constitute a simultaneous test; intervals containing zero do not establish equivalence or exact additivity.

6 Computational expenditure as part of the design

6.1 Definition of the computational workload

A time comparison is meaningful only after defining the computational unit. For a collection of frozen policies under one financial model, approximate-value construction and the common martingale can be shared. Charging this construction once per policy overstates its cost; omitting it understates the cost. A single-policy workload can have a different ranking from a repeated multi-policy workload.

The timed unit should state whether it includes simulation, deterministic approximation, conditional integration, martingale evaluation, policy realisation, estimator construction and numerical checks. Previous training, imports, data loading and file output should be reported separately if excluded. Warm-up and allocation pilots are additional expenses, not free inputs to an end-to-end valuation. Record elapsed and CPU time and the relevant hardware and thread configuration.

6.2 Path allocation before evaluation

I allocate paths using median pilot times t_1, t_2 at two path counts $n_1 < n_2$:

$$b_T = \frac{t_2 - t_1}{n_2 - n_1}, \qquad a_T = \max(0, t_1 - b_T n_1), \qquad \widehat{T}(n) = a_T + b_T n. \tag{25}$$

For target B and path increment q, choose

$$n(B) = q \left\lfloor \frac{B - a_T}{b_T q} \right\rfloor, \tag{26}$$

subject to prescribed limits. A non-positive fitted slope or infeasible minimum is a design failure. The unclipped intercept, timing dispersion and any cap must be retained. I fix all allocations before drawing the independent evaluation streams and retain each prescribed run irrespective of whether its confidence bound becomes favourable.

The affine model predicts time; it does not enforce it. Extrapolation beyond the pilot range, memory effects and host variability can create substantial discrepancies. I compare actual time with target time and retain overruns. A strict equal-cost ranking requires adequate timing calibration or a separately justified comparison on an observed cost frontier. Nominally equal targets alone do not establish it.

6.3 Allocating repetitions of a fixed training comparison

The value of a realised policy and the mean value of a training procedure are different estimands. To study the latter, I fix the financial specification, two training procedures, their training sample sizes and the common evaluation martingale. Let D contain the inputs needed to fit one pair of policies, with a prescribed joint law ν. This law may couple the fits through shared training paths, as in the retraining comparison above. Independence is required between successive pairs and their fresh evaluation paths, not between the two fits within a pair. Increasing the number of pairs repeats the same procedures; it does not enlarge the training sample of an individual fit.

For a complete evaluation path X, define the quoted adjusted gain

$$Z(D, X) = \frac{L_{\pi_1(D)}(M) - L_{\pi_0(D)}(M)}{b}, \qquad \delta(D) = \mathbb{E}[Z \mid D], \qquad \sigma_D^2 = \mathrm{Var}(Z \mid D). \tag{27}$$

Expectations in this subsection include both training and evaluation randomness. Under independent evaluation and the martingale condition, $\delta(D) = \{V_{\pi_1(D)} - V_{\pi_0(D)}\}/b$ almost surely. Assume Z has a finite second moment under the joint sampling law. The fixed target and its two variance components are

$$\mu = \mathbb{E}[\delta(D)], \qquad \mathcal{V}_{\mathrm{tr}} = \mathrm{Var}\{\delta(D)\}, \qquad \mathcal{V}_{\mathrm{ev}} = \mathbb{E}[\sigma_D^2]. \tag{28}$$

Let $\mathcal{P}$ contain all preceding evidence used to choose positive integer counts m and n, before any new outcome is drawn. For each of m independent training pairs, I average n fresh evaluation gains to obtain W_r and then set $\widehat{\mu} = m^{-1} \sum_{r=1}^{m} W_r$. The product-space construction in Appendix B gives

$$\mathbb{E}[\widehat{\mu} \mid \mathcal{P}] = \mu, \qquad \mathrm{Var}(\widehat{\mu} \mid \mathcal{P}) = \frac{\mathcal{V}_{\mathrm{tr}}}{m} + \frac{\mathcal{V}_{\mathrm{ev}}}{mn}. \tag{29}$$

I obtain this decomposition by conditioning first on the fitted pair and then averaging over independent pairs. Its purpose is to distinguish variation between fitted policies from the Monte Carlo error remaining after those policies are fixed. Additional paths within a pair reduce only the second component.

For an illustrative deterministic CPU budget $B_{\text{cpu}} = c_0 + m(c_{\text{tr}} + c_{\text{ev}}n)$, the unconstrained continuous relaxation gives

$$n^* = \sqrt{\frac{\mathcal{V}_{\text{ev}} c_{\text{tr}}}{\mathcal{V}_{\text{tr}} c_{\text{ev}}}}, \qquad m^* = \frac{B_{\text{cpu}} - c_0}{c_{\text{tr}} + c_{\text{ev}} n^*}, \tag{30}$$

when both variance components and both marginal costs are positive and $B_{\text{cpu}} > c_0$. Substituting the budget constraint into (29) leaves the terms $\mathcal{V}_{\text{tr}} c_{\text{ev}} n + \mathcal{V}_{\text{ev}} c_{\text{tr}}/n$ as the only terms depending on n; differentiation yields (30). Zero components, integer counts and minimum sample sizes require boundary comparisons instead. Actual allocation requires estimated components and measured costs, with their uncertainty retained. This calculation supplies neither a verified CPU limit nor evidence that a particular allocation is best.

The counts may depend on earlier evidence while remaining fixed for the new experiment. Stopping the new experiment when its own accumulated cost crosses a threshold is a different sampling rule. Stochastic costs are part of the efficiency problem studied by Glynn and Whitt (1992); Appendix B gives a finite example in which an outcome-dependent cost rule biases the resulting sample mean. I retain incomplete workloads as incomplete rather than applying (29) to a selectively completed subset.

6.4 Approximation and sampling serve different purposes

A more detailed approximate value can tighten the dual expectation while making each outer path more expensive. Under a time constraint this can reduce the available sample size and increase the sampling radius in (20). Report the mean gap, its standard error, the upper limit and actual runtime separately. A finer deterministic grid is not automatically the best use of a limited evaluation budget.

For independent-method evaluations, use (23); for common-path evaluations, retain (22). Report all retained policies and repetitions. Descriptive averages or medians across related cases are not independent observations of a population effect. Neither a tighter dual construction nor a larger outer sample changes a frozen policy's expected reward. Comparing these expenditures with extra policy training is a distinct resource-allocation problem.

7 Conditional computational evidence

7.1 Specifications, procedures and estimands

The investigation uses thirteen related development specifications for one SPY expiry, anchored to observations from a short intraday window on 10 September 2026. They are neither independent trading days nor a representative market sample. Each specification retains a one-state diffusion, deterministic funding and a capped discrete cash event. The original calendar contains 27 daily exercise rights. I treat these inputs as defining conditional numerical problems. The calculations do not establish that the retained funding, cash or quote conventions describe contemporaneous tradable opportunities.

The original design crosses 5,000 and 20,000 training paths with polynomial degrees two and four. Four repetitions per case give 52 policies per configuration and 208 policies overall. The smaller training sample is a prefix of the larger sample in this design. Independent outer paths evaluate the frozen rules. Later experiments draw independent pairs of training samples to study procedure-level variation; the policies within each pair share evaluation paths. I distinguish a change in the expected reward of a policy, a change in the precision of its assessment, and a change in expenditure. These distinctions determine the comparisons below.

All monetary results are USD per share in premium-quote units. The tables report the reviewed derived summaries; exact financial input bundles are excluded from the public edition. The observation-admission study in Section 8 remains a separate assessment of empirical validity.

7.2 Bound tightness and terminal policy gains

Table 1 compares the slope martingale with coarse and fine approximate-value constructions while keeping each exercise policy unchanged. The fine construction reduces the mean estimated gap by 91.42 to 95.23 per cent relative to the slope construction. Its improvement over the coarse construction is substantially smaller. By equation (8), changing the martingale changes the dual slack and leaves the expected reward of the policy fixed. The comparison measures an improvement in loss assessment, rather than additional reward earned by the holder. The averages describe related retained specifications and do not estimate a population effect across markets.

Table 1: Expected-gap estimates for unchanged exercise policies

Training paths	Degree	Slope	Coarse value	Fine value	Reduction (per cent)
5,000	2	0.652833	0.051285	0.051125	92.17
5,000	4	0.703839	0.060522	0.060364	91.42
20,000	2	0.624587	0.037099	0.036941	94.09
20,000	4	0.506690	0.024304	0.024150	95.23

Each entry averages 52 policies. Gaps are USD per share in premium-quote units and contain both policy loss and dual slack. Reduction compares the fine value construction with the slope construction.

The terminal experiment adds ninety minutes to the original endpoint, with no intervening cash event and a constant settlement threshold. Earlier decisions are retained and a surviving policy waits until the later endpoint. Under the assumptions of the terminal-adjustment proposition, conditional convexity gives a non-negative expected gain even though individual realised payoffs can decrease. Table 2 uses the integrated original-clock comparison in (18), which represents the same expanded opportunity set under endpoint dominance. Its common upper term cancels, so the paired expected-gap reduction identifies a policy-value gain. These integrated gaps differ from both the direct later-payoff diagnostic and the original-calendar gaps in Table 1.

Table 2: Terminal adjustment under a common upper payoff

Training paths	Degree	Unadjusted gap	Adjusted gap	Policy gain
5,000	2	0.06111782	0.05009106	0.01102676
5,000	4	0.07063052	0.06023823	0.01039229
20,000	2	0.04703011	0.03593234	0.01109777
20,000	4	0.03441110	0.02377610	0.01063500

Means across 52 original policies per configuration, in premium-quote units. Both instructions are assessed on the expanded calendar. The common upper term cancels in their paired difference.

For the 20,000-path degree-four configuration, the gap estimate falls from 0.03441110 to 0.02377610, giving a paired gain of 0.01063500. This conclusion concerns a specified feasible change in the terminal instruction. Revising the earlier continuation regressions is a different intervention. In the separate matched terminal-target retraining comparison, the 208 pointwise intervals comprise 81 positive intervals, 108 negative intervals and 19 intervals containing zero. These dependent comparisons do not establish a simultaneous policy ranking. They show why an analytically justified change in a terminal instruction does not imply that every finite-sample refit will improve its preceding decisions.

7.3 Variation between fitted policies

The repeated-training comparison fixes degree four and compares policies fitted using 5,000 and 20,000 paths. It draws eight independent training pairs per specification; the two training samples within each pair are independent in this experiment. A common fine value martingale and 80,000 common evaluation paths estimate their adjusted gain. Table 3 reports the resulting mean and pointwise Student interval across the eight pair means, together with the estimated contribution of within-pair evaluation noise to the variance of the mean estimator.

Table 3: Mean gains across independent training pairs

Case	Mean gain	Pointwise interval	Evaluation share (per cent)
0	0.03203	[0.00922, 0.05484]	0.071
1	0.02721	[0.00193, 0.05249]	0.046
2	0.02941	[−0.00635, 0.06517]	0.029
3	0.04352	[0.01747, 0.06958]	0.077
4	0.03362	[0.01262, 0.05462]	0.123
5	0.03872	[0.02509, 0.05236]	0.284
6	0.03173	[0.01776, 0.04570]	0.152
7	0.03283	[0.01873, 0.04694]	0.179
8	0.02892	[0.01123, 0.04661]	0.144
9	0.01419	[−0.00237, 0.03076]	0.107
10	0.05397	[−0.00186, 0.10980]	0.014
11	0.03105	[0.02017, 0.04194]	0.267
12	0.03255	[0.00226, 0.06285]	0.056

Eight pairs per case; 80,000 common evaluation paths per pair. Intervals use a pointwise 95 per cent Student approximation across pair means. Finite-sample coverage is unverified; the thirteen intervals are not simultaneous.

Every estimated mean is positive, but the intervals for cases 2, 9 and 10 contain zero. Eight of the 104 individual pair gains are negative. Omitting repetitions one at a time leaves all reduced-sample means positive, while the positive-interval conclusions for cases 1 and 12 depend on the omission. These diagnostics qualify the interpretation of eight repetitions; they do not replace the original estimates. The reported intervals are pointwise approximations, and their finite-sample coverage has not been validated.

The estimated evaluation shares range from 0.014 to 0.284 per cent. Under equation (29), extra evaluation paths reduce only the within-pair component, whereas additional independent pairs reduce both components. The small estimated shares therefore motivate shifting expenditure towards more repetitions of the fixed training procedures. They are estimated variance components, not known population quantities, and do not establish a preferred allocation without a cost comparison.

7.4 Independent allocation and failed budget transfer

A new pilot fits 32 independent pairs per case and evaluates each pair on 4,096 paths. Separate timing measurements cover 128, 256, 512, 1,024, 2,048 and 4,096 evaluation paths. The reference uses 64 pairs and 4,096 paths per pair. The selector minimises estimated variance over affordable integer counts from 64 to 1,024 and the measured path grid, using the pilot's predicted reference CPU budget. A non-positive estimated component invokes the reference allocation. The allocation is fixed before new main draws; a comparison is not dropped because its realised cost or gain is unfavourable.

The independent main arms complete 832 reference pairs and 4,198 selected pairs. The selected allocations use 256 evaluation paths in twelve cases and 128 in case 12. The latter is a measured-grid boundary choice and gives no evidence about smaller inner counts. Table 4

reports the descriptive ratios. Main variance ratios range from 0.157 to 0.507 and variance-times-CPU ratios from 0.155 to 0.485. These ratios have no confidence intervals. Several are less favourable than predicted, particularly for cases 6 and 11.

Table 4: Independent allocation comparison

Case	Pairs	Paths	Pilot variance	Main variance	CPU	Variance times CPU
0	306	256	0.237	0.204	0.962	0.197
1	352	256	0.230	0.311	1.078	0.336
2	327	256	0.238	0.268	1.041	0.279
3	332	256	0.224	0.179	1.026	0.184
4	318	256	0.272	0.311	0.970	0.302
5	318	256	0.262	0.331	1.042	0.345
6	338	256	0.253	0.486	0.998	0.485
7	335	256	0.220	0.157	0.991	0.155
8	333	256	0.246	0.266	1.001	0.266
9	320	256	0.261	0.277	1.011	0.280
10	279	256	0.255	0.356	0.924	0.329
11	253	256	0.327	0.507	0.659	0.334
12	387	128	0.238	0.247	0.930	0.230

Counts describe the selected allocation. Ratios are selected relative to the independent reference, which uses 64 pairs and 4,096 paths. Variance refers to the estimated variance of the mean gain. Ratios are descriptive and have no ratio confidence intervals.

The additional independent repetitions give more information about the procedures' average gain than the eight-pair experiment. Both arms have positive pointwise intervals in every case (Table 5). This remains conditional inference across training repetitions, not independent confirmation from new market observations. Increasing the number of fitted pairs improves estimation of the fixed procedures' mean difference; it does not improve each fitted policy by increasing its own training sample.

Table 5: Training-procedure gains under the two allocations

Case	Reference mean	Reference interval	Selected mean	Selected interval
0	0.03407	[0.02661, 0.04153]	0.02988	[0.02656, 0.03319]
1	0.02515	[0.01936, 0.03094]	0.03296	[0.02978, 0.03614]
2	0.03214	[0.02590, 0.03838]	0.03354	[0.03035, 0.03672]
3	0.03219	[0.02472, 0.03966]	0.02970	[0.02658, 0.03281]
4	0.03112	[0.02517, 0.03707]	0.03028	[0.02701, 0.03355]
5	0.02989	[0.02381, 0.03598]	0.03136	[0.02791, 0.03481]
6	0.02916	[0.02361, 0.03472]	0.03502	[0.03121, 0.03883]
7	0.03858	[0.02998, 0.04718]	0.03161	[0.02826, 0.03497]
8	0.02732	[0.02058, 0.03406]	0.03141	[0.02799, 0.03483]
9	0.03380	[0.02761, 0.04000]	0.03278	[0.02957, 0.03600]
10	0.03578	[0.02878, 0.04279]	0.04138	[0.03726, 0.04549]
11	0.03114	[0.02531, 0.03698]	0.03381	[0.02972, 0.03791]
12	0.03329	[0.02667, 0.03991]	0.03304	[0.02980, 0.03628]

Premium-quote units. Intervals use a pointwise 95 per cent Student approximation over independent pair means. Positive intervals concern the fixed conditional specifications, not independent market confirmations.

Every one of the 26 main arms falls below the prescribed forecast-tolerance interval. Actual CPU divided by the pilot's reference budget ranges from 0.392 to 0.811, against the interval $[0.9, 1.1]$. Selected-to-reference actual CPU ratios are closer to one, ranging from 0.659 to 1.078, but this does not establish equality of realised expenditure. I retain the fixed allocations instead of using the apparent surplus to add observations after inspecting their outcomes.

Table 6: Predicted and actual allocation CPU expenditure

Component	Reference predicted	Reference actual	Selected predicted	Selected actual
Setup	2.333	1.104	2.333	1.060
Training	75.569	34.814	366.801	166.359
Evaluation	595.503	270.059	303.460	129.534

Aggregate seconds. Component predictions retain the pilot observation supplying the median total pair cost. The separate pilot expense is excluded from this table and reported in the text.

Table 6 decomposes the aggregate numerical charges. Predicted totals of 673.405 and 672.594 CPU seconds become 305.977 and 296.953 seconds for the reference and selected arms. Predictions retain the training and evaluation components of the pilot observation supplying the median total pair cost. Summing separate component medians would generally change the forecast. Both training and evaluation become cheaper; setup explains less than one per cent of either shortfall. Recorded software and requested thread settings agree across stages, but the effective native thread counts and processor conditions were not measured for this earlier comparison. I cannot identify the cause of the change.

The pilot additionally consumes 457.980 process CPU seconds and 639.279 elapsed seconds. This is a separate expense in a one-off application. The evidence supports improved estimated precision under the retained specifications, while leaving equal-realised-cost superiority and budget transfer unestablished. A subsequent experiment assesses new forecasts; it cannot retrospectively repair these original predictions.

7.5 A prespecified computational follow-up

I selected cases 5, 6 and 10 because earlier evidence exposed uncertainty about variance components or sensitivity to the cost scenario. Each case has three candidate allocations, using 128, 256 and 4,096 evaluation paths. The count of independent training pairs is the largest affordable integer under both retained mean-cost scenarios. A prespecified rule minimises worst relative variance over a declared enclosure of the asymptotic component-ratio set, as derived in Appendix C. The rule selects 256 paths in all three cases. The set's coverage and the cost scenarios remain assumptions of that decision.

All nine arms are executed, completing 3,978 training pairs, with four warm-up pairs recorded separately. The earlier information determines the design and cases, while the new simulation outcomes do not determine their own counts. This is prospective computational evidence conditional on the preceding research and fixed financial specifications. It supplies no new market observations.

Table 7: Prespecified computational allocation follow-up

Case	Pairs 128	Pairs 256	Pairs 4,096	Variance 128	Variance 256	CPU 256	Variance times CPU 256
5	776	687	91	0.169888	0.167421	1.510736	0.252928
6	674	632	141	0.396693	0.340606	0.901768	0.307147
10	438	399	140	0.364378	0.330904	0.655844	0.217021

Column suffixes give evaluation paths per pair. Ratios use the fresh within-case 4,096-path reference. The prespecified choice is 256 paths. Ratios and comparisons between 128 and 256 paths remain descriptive; realised costs differ.

The selected 256-path variance estimates are 16.7, 34.1 and 33.1 per cent of their fresh within-case reference estimates (Table 7). Their variance-times-CPU ratios are also below one. Actual expenditures differ, so these point estimates do not establish an improvement at exactly

equal realised cost. The 128-versus-256 point comparison favours 256 in each case but is outside the registered inferential family.

The registered family contains twelve contrasts: six mean differences and six differences between the estimated variances of the complete-sample means. Each smaller-path arm is compared with its independent 4,096-path reference. Intervals use $\widehat{\Delta} \pm (13/4)\widehat{s}$. The cutoff supplies an asymptotic union-bound calibration for the family; shared references do not require independence between the contrasts. Appendix C states the moment assumptions and the variance-influence calculation. Neither exact arithmetic in the decision rule nor a conservative normal cutoff establishes finite-sample coverage.

Table 8: Registered mean and estimator-variance difference intervals

Case	Paths	Mean difference	Variance difference
5	128	$[-0.009932236, 0.01029046]$	$[-1.091314 \times 10^{-5}, -2.823286 \times 10^{-6}]$
5	256	$[-0.009629916, 0.01057145]$	$[-1.093775 \times 10^{-5}, -2.839513 \times 10^{-6}]$
6	128	$[-0.005816441, 0.009489616]$	$[-3.959426 \times 10^{-6}, -8.309205 \times 10^{-7}]$
6	256	$[-0.005216458, 0.009779123]$	$[-4.164466 \times 10^{-6}, -1.071226 \times 10^{-6}]$
10	128	$[-0.01285611, 0.009544336]$	$[-1.018491 \times 10^{-5}, -8.808303 \times 10^{-7}]$
10	256	$[-0.01311862, 0.009005322]$	$[-1.041016 \times 10^{-5}, -1.23835 \times 10^{-6}]$

Each contrast subtracts the independent 4,096-path reference. Mean differences use premium-quote units and variance differences their square. The common cutoff 13/4 supplies asymptotic family calibration for twelve contrasts; finite-sample coverage is unverified.

All six mean intervals include zero, which is compatible with the common target but does not establish equivalence. All six variance-difference intervals have negative upper endpoints, supporting the precision comparison under the specified asymptotic procedure. Fourth-moment concentration qualifies that interpretation: one pair supplies 63.98 per cent of the squared variance-influence sum in case 5's 256-path arm, and the largest contribution is 49.84 per cent in case 10's reference. These are shares of estimated uncertainty about variance, not shares of the variance itself. No pair is removed.

All nine arms are within their original numerical budgets, but none meets the ten per cent forecast tolerance. Actual-to-forecast ratios range from 0.514162 to 0.873429. The primary numerical charge is 349.644 CPU seconds against a combined forecast of 487.779 seconds. This second comparison strengthens the conditional precision evidence while leaving expenditure prediction unresolved at that stage.

7.6 Local CPU calibration

The final timing experiment asks whether local measurements predict those nine complete allocations more accurately. Two passes execute fixed counts in opposite orders, giving 7,956 validation pairs, 576 calibration pairs and four warm-up pairs. At each pass's start, sixteen calibration pairs per arm determine a pass-start forecast. Another sixteen pairs immediately before each arm determine an immediate forecast. Both extrapolate mean pair cost and include a fresh approximation construction. The historical forecast is unchanged. Reversal changes eight positions; the middle arm retains its position.

For actual numerical CPU C and forecast $\widehat{C}$, I assess $|C - \widehat{C}|/\widehat{C}$, averaging equally across eighteen arms. The tolerance requires $C/\widehat{C} \in [0.9, 1.1]$. A separate upper allowance is $1.1\widehat{C}$, without a statistical prediction-coverage interpretation. Table 9 preserves all eighteen outcomes.

Local calibration reduces mean absolute relative error from 29.065 per cent to 3.669 per cent at pass start and 2.841 per cent immediately before execution. The immediate forecast meets the tolerance in seventeen of eighteen arms. Its remaining failure has an actual-to-forecast ratio of 1.2261 and exceeds the original numerical budget. Every point forecast classifies all eighteen arms as affordable, although only seventeen are. Thus the improvement in average prediction

Table 9: Historical and locally calibrated CPU forecasts

Forecast	Within ten per cent	Within upper allowance	Mean absolute error (per cent of forecast)	Actual / forecast range
Historical	1/18	18/18	29.065	[0.5154, 1.0782]
Pass start	16/18	16/18	3.669	[0.9237, 1.2081]
Immediate	17/18	17/18	2.841	[0.9576, 1.2261]

Each forecast is assessed on the same eighteen complete validation charges. The error denominator is the forecast; arms receive equal weight. The upper allowance is 1.1 times the forecast and has no statistical coverage guarantee.

does not establish reliable budget protection.

The historical upper allowance contains all eighteen costs, but classifies none of the planned allocations as affordable. Pass-start and immediate allowances classify eighteen and seventeen arms as affordable, respectively, with one false affordability declaration each. Tolerance, upper-allowance exceedance and affordability answer different planning questions. Covering all observed costs is insufficient if the resulting allowance makes every planned allocation infeasible.

Validation uses 700.274 numerical CPU seconds. Pass-start and immediate calibration add 44.623 and 44.323 seconds, and warm-up adds 0.332 seconds. Total numerical CPU is 789.552 seconds; whole-process CPU is 811.658 seconds and elapsed time is 842.795 seconds. Process timing includes in-run recording and control operations but excludes interpreter startup, pre-launch verification and post-run manifest hashing. Charging either calibration scheme alone retains seventeen affordable arms. Charging both, as executed here, leaves fourteen. The latter count describes this two-scheme investigation rather than deployment of immediate calibration alone.

Both passes share one operating session. Fresh simulation streams do not make processor conditions independent and identically distributed, and reverse order does not identify a causal order effect. Numerical kernels are unchanged, but this experiment retains compact verification records rather than serialising every fitted coefficient. Whole-process timing therefore belongs to that recording workflow. I use these runs to assess cost forecasts, without treating them as additional inference about financial gains or a guarantee for another processor.

8 Financial validation and interpretation

A numerical stopping problem can be well defined while its market interpretation is unresolved. Empirical application requires dated contract terms, a justified ownership state, cash information available at the valuation cutoff, and funding appropriate to the payment obligations. Event and source timestamps may describe different objects. Simultaneous records can require an explicit ordering rule; absent or stale inputs must not silently become valid observations.

I registered a prospective primary sample of fifteen matched observations over five sessions, 21–25 September 2026. No observation meets the retained admission requirements. I report no performance estimate for that sample. Subsequent access checks do not supply the missing evidence needed to admit it. In particular, the retrieved curve information does not provide a complete verified set of dated funding inputs, and a partially identified curve cannot be treated as a completed bootstrap. The original empirical criterion remains unmet. The narrower completed scope is a conditional computational study; I have not relaxed the original screen to create a successful validation sample.

Joint call and put compatibility under a shared state, cash specification, funding convention and volatility is an informative diagnostic. Separate unconstrained calibrations can remove that restriction. Compatibility with bid–ask intervals is still not identification of the correct model or an executable arbitrage. A model that fails a renewed quote diagnostic after a sensitivity change can remain a conditional mathematical case, but must not be reclassified as an accepted empirical specification because its policy loss is small.

A quoted spread offers a monetary scale. It is not automatically the cost of exercise, hedging or implementation. An upper loss limit below the spread establishes only that conditional scale comparison. A limit above it leaves the inequality unestablished; it does not prove that true loss exceeds the spread. The results above accordingly concern model rewards and computation, without a trading-profitability claim.

Additional historical quotes with documented sampling conventions could support a new empirical design. Such a design would need its own observation rule, dated cash and funding information, and independent evaluation. It would not retrospectively convert the present failed admission exercise into a successful prospective test. Neither fresh simulation seeds nor additional training repetitions replace new financial observations.

9 Conclusions and data availability

The completed study connects loss assessment, policy comparison and computational allocation in one funded cash-event stopping model. Each comparison retains an explicit financial problem, exercise instruction or sampling target so that its resulting change can be interpreted.

The first comparison separates measurement from action. Analytically centred approximate values substantially reduce estimated dual gaps for unchanged policies. A common upper payoff then isolates the expected gain from a feasible terminal adjustment. Finite-sample retraining can still worsen individual fitted rules, so a favourable terminal argument cannot justify every earlier regression decision. The relevant object is the policy's reward under the stated exercise opportunities.

The second comparison concerns the precision of a training-procedure contrast. Conditional total variance identifies the variation between fitted pairs and the evaluation noise remaining within each pair. Reallocating computation towards independent pairs reduces the retained variance estimates. The prespecified follow-up supports that conclusion under its asymptotic contrast procedure, while influential fourth moments limit finite-sample confidence. More precise estimation of a fixed training comparison is distinct from improving each policy or establishing population-optimal allocation.

The third comparison shows why the cost model belongs inside the research design. Historical CPU forecasts fail to transfer even when the numerical procedures remain fixed. Local calibration improves prediction for the measured session but costs computation and does not eliminate a budget overrun. A favourable variance-times-CPU ratio therefore cannot substitute for a validated expenditure forecast or an end-to-end cost accounting.

The probability arguments specify which conditioning makes these conclusions coherent. Independent training information preserves the evaluation martingale, fixed fresh counts preserve the repeated-training estimand, and predictable compensation identifies the correction needed when conditional centring fails. These arguments explain the implemented comparisons and their limits. They do not establish theoretical novelty, exact small-sample coverage or validity of the maintained financial inputs.

I publish the derived numerical findings, the self-contained LaTeX source and the relevant numerical implementation. The accompanying notebooks execute analytical examples and inspect the released result summaries. A separate replication interface runs the principal kernels on explicitly stipulated mathematical inputs. Those examples are not observed market data and do not reproduce the financial result tables. Exact replay of the retained financial experiments requires their original licensed inputs and evidence records, which remain private; independently supplied authorised inputs define new conditional experiments. Raw provider exports and observation-level subscription inputs are excluded from this release. The original market-validation criterion remains unsatisfied, and any extension to additional contracts, state variables or operating environments requires further evidence.

A Conditional drift and the interpretation of a gap

The martingale requirement is conditional. Zero average increments do not suffice. This distinction is established in the dual literature: Belomestny et al. (2009) preserve the martingale property through their construction, while Conditions A and B of Cline (2026) state conditional restrictions on projection residuals. I apply predictable compensation to express the consequence of a violation in the units of policy loss and paired policy gains. The purpose is to measure the error introduced by imperfect conditional centring. Subtracting accumulated conditional increments produces a martingale; optional sampling then isolates the part of the apparent gap that cannot be attributed to policy loss. The following argument is an elementary consequence of the cited dual framework, not a claim of a new theorem or an original proof.

A.1 A correction for an adapted penalty

Fix the trained functions and let $\mathcal{F}_j$ denote the evaluation information at exercise right j. Suppose the adapted rewards H_j and an adapted penalty A_j are integrable, $A_0 = 0$, and exercise is allowed at $1, \dots, J$. Define

$$\mu_j = \mathbb{E}[A_j - A_{j-1} \mid \mathcal{F}_{j-1}], \qquad B_j = \sum_{k=1}^{j} \mu_k, \qquad M_j = A_j - B_j, \quad B_0 = 0. \tag{31}$$

Write $V^* = \sup_\tau \mathbb{E}H_\tau$ and $V_\pi = \mathbb{E}H_{\tau_\pi}$, with the supremum taken over admissible stopping times. Then M is an integrable martingale starting at zero. For an admissible policy π, write

$$U(A) = \max_{1 \le j \le J} (H_j - A_j), \qquad G_\pi(A) = U(A) - H_{\tau_\pi} + A_{\tau_\pi}, \tag{32}$$

$$D_\pi = \max_{1 \le j \le J} B_j - B_{\tau_\pi}, \qquad \eta = \sum_{k=2}^{J} \mathbb{E}|\mu_k|. \tag{33}$$

The gap and D_π are non-negative pathwise. Nevertheless, $U(A)$ need not have an expectation above the optimal value.

Proposition 4 (Compensated loss bound)**.** *Under these assumptions,*

$$0 \le V^* - V_\pi \le \mathbb{E}[G_\pi(A) + D_\pi] \le \mathbb{E}[G_\pi(A)] + \eta. \tag{34}$$

Proof. For every admissible stopping time τ, $H_\tau \le U(A) + A_\tau$. Optional sampling applied to $M = A - B$ gives $\mathbb{E}A_\tau = \mathbb{E}B_\tau$, and likewise for τ_π. Therefore

$$\mathbb{E}H_\tau - V_\pi \le \mathbb{E}G_\pi(A) + \mathbb{E}(B_\tau - B_{\tau_\pi}) \le \mathbb{E}G_\pi(A) + \mathbb{E}D_\pi.$$

Taking the supremum over τ gives the first upper bound. The difference between any two values of B at rights $1, \dots, J$ is bounded by $\sum_{k=2}^{J} |\mu_k|$. This gives the second upper bound. Non-negativity of the loss follows from admissibility of π. □

The sum starts at two because B_1 is common to every eligible stopping right and cancels from the policy-loss comparison. If exercise at zero is included, the corresponding sum starts at one. If $\|\mu_k\|_2$ is finite, then $\mathbb{E}|\mu_k| \le \|\mu_k\|_2$ under the evaluation probability measure. Certified bounds on these norms thus suffice for this expectation bound without a uniform statewise bound. Neither finite regression residuals nor a sample estimate of a norm supplies such a certificate automatically. Division by the positive premium discount factor b converts the entire bound, including the drift allowance, into quoted premium units.

A.2 Paired policies

For two policies evaluated with the same adapted penalty, the pathwise cancellation of the upper payoff remains exact. Its expectation satisfies

$$\mathbb{E}[G_\pi(A) - G_\rho(A)] = V_\rho - V_\pi + \mathbb{E}[B_{\tau_\pi} - B_{\tau_\rho}]. \tag{35}$$

This follows by subtracting the definitions of the gaps and using $\mathbb{E}A_\tau = \mathbb{E}B_\tau$. The absolute value of the last term is at most η. Common-path evaluation therefore removes the common upper payoff but does not remove conditional-drift bias. This matters when comparing policies that stop in different states: the same residual process can be selected differently by their stopping times.

A.3 A finite analytical example

Let Z take values -1 and 1 with equal probability and become known at right one. There are two exercise rights and no additional information is needed at right two. Set $H_1 = 1$, $H_2 = 1+Z$, $A_1 = 0$ and $A_2 = Z$. These are stipulated mathematical quantities, not observations or fitted market inputs. Both unconditional increment means are zero, but $\mu_2 = Z$.

A policy π that always waits has value 1. The optimal policy stops at one when $Z = -1$ and waits otherwise, with value $3/2$. Yet $U(A) = 1$ and $G_\pi(A) = 0$ in both states. Thus a non-negative, identically zero apparent gap coexists with a positive policy loss. Here $D_\pi = (-Z)^+$, so $\mathbb{E}D_\pi = 1/2$ and the first corrected bound is attained. The coarser allowance is $\eta = 1$.

The optimal policy also has a zero apparent gap in both states. Consequently the paired gap difference is zero although the true gain is $1/2$. The compensator term in (35) is $-1/2$, accounting for the discrepancy. The example illustrates the conditional requirement; I make no claim of priority for this construction.

A.4 Numerical and inferential scope

For the analytically centred functions used in this paper, $\mu_j = 0$ in exact arithmetic when the stated transition and retained function are integrated exactly. The correction concerns substitutions such as approximate conditional integration or projection. It is not evidence of a defect in the retained construction, and it does not itself certify floating-point error.

If an evaluation procedure supplies an upper confidence limit for $\mathbb{E}G_\pi(A)$ and a deterministic certified bound on η, their sum has the same coverage for policy loss. If the drift allowance is estimated, its uncertainty requires separate control, for example by simultaneous limits with a declared combined error probability. An asymptotic limit for the gap remains asymptotic after the correction. Testing unconditional sample means cannot replace the conditional requirement.

The accompanying `verification/conditional_drift.py` enumerates stopping policies on finite trees using rational arithmetic. It checks the displayed example, the compensated inequality and the paired identity against an independently evaluated dynamic programme. These calculations validate the finite implementation of the argument. They neither establish theoretical novelty nor supply empirical findings.

B Training information and conditional allocation

B.1 The sampling space and the learned stopping rule

I make the probability construction explicit because a fitted stopping rule uses information acquired before its evaluation path begins. The financial specification, training procedures, their within-pair coupling and the retained approximate values are fixed throughout this construction. Let λ be the law of one complete evaluation path induced by the maintained pricing measure $\mathbb{Q}$, and let ν be the joint law of one training pair. Let the preceding record have law Π and generate $\mathcal{P}$. On standard Borel record, training and path spaces, an ideal simulation law is

$$\mathbb{P} = \Pi \otimes \bigotimes_{r\geq 1} \left(\nu \otimes \bigotimes_{i\geq 1} \lambda \right). \tag{36}$$

This construction supplies independent rows $(D_r, (X_{ri})_{i\geq 1})$ independent of the preceding record. Within a row the same fitted pair is used for every path. Conditional on $\mathcal{P}$, the finite positive counts $m = m(\mathcal{P})$ and $n = n(\mathcal{P})$ select an initial rectangle of this fresh array; they do not alter its product law. These independence statements are assumptions of the sampling model, not consequences of assigning different names to random-number streams.

Each training rule must be measurable in its input data. In a finite regression algorithm, measurable state filters, arithmetic on their declared domains, specified tie rules and finitely many safeguard branches preserve measurability. For a real design matrix A, its Moore–Penrose inverse is Borel measurable because

$$A^{+} = \lim_{\varepsilon\downarrow 0} (A^{\mathsf{T}}A + \varepsilon I)^{-1} A^{\mathsf{T}}$$

is a pointwise limit of continuous maps. A finite, explicitly specified rank or degree-selection rule therefore permits backward induction to produce jointly measurable continuation functions of training data and state. This is a requirement on the implemented rule, including its exceptional-input behaviour; it is not an assertion that an unspecified numerical routine has this property.

For one fresh path, write $\mathcal{F}_j$ for its information through exercise right j and enlarge initially to

$$\mathcal{G}_j = \mathcal{P} \vee \sigma(D) \vee \mathcal{F}_j.$$

The fitted policy is a bounded $\mathcal{G}_j$-stopping time because its parameters are initially available and its exercise predicates use no future path state. The training pair and preceding record are independent of the complete fresh path. Consequently, for the retained integrable value function $v_{\kappa(j)}$ at exercise right j,

$$\mathbb{E}_{\mathbb{P}}[v_{\kappa(j)}(Y_j) \mid \mathcal{G}_{j-1}] = \mathbb{E}_{\lambda}[v_{\kappa(j)}(Y_j) \mid \mathcal{F}_{j-1}]. \tag{37}$$

The same initial-enlargement argument applies to $\mathcal{H}_k$ at every propagation node in (10); restricting its martingale to the exercise rights then preserves the required conditional property. Integrability at the finitely many rights permits bounded-time optional sampling, conditionally on $\mathcal{P}$ and D. It gives $\mathbb{E}[M_{\tau_{\pi_a(D)}} \mid \mathcal{P}, D] = 0$ for $a = 0, 1$ and hence the expression for $\delta(D)$ below (27). I use this argument to justify the training-averaged policy comparison, applying the approximate-value construction of Haugh and Kogan (2004) to the regression policies of Longstaff and Schwartz (2001).

Equation (37) fixes the functions v_k. If the martingale functions are also trained, their inputs must be included in the initial training information and their conditional expectations must integrate the realised functions. Integrating an average function in their place need not centre the realised increment. Additional joint integrability is then required. The initial enlargement cannot simply be omitted because the parameters were fitted before evaluation.

B.2 Variance, estimation and the scope of conditioning

Under the finite-second-moment assumption for Z, paths within a row are conditionally independent given D_r with mean $\delta(D_r)$ and variance $\sigma^2_{D_r}$. Conditional total variance yields

$$\mathbb{E}[W_r \mid \mathcal{P}] = \mu, \qquad \mathrm{Var}(W_r \mid \mathcal{P}) = \mathcal{V}_{\mathrm{tr}} + \mathcal{V}_{\mathrm{ev}}/n.$$

Distinct rows are conditionally independent, which proves (29). Individual gains within a row generally are not independent after training is averaged out: two different gains have covariance $\mathcal{V}_{\mathrm{tr}}$. Treating all mn gains as independent would therefore remove the training component incorrectly.

For $m \geq 2$, let S^2_W be the usual sample variance of the m row means. Then S^2_W/m is conditionally unbiased for the variance in (29). Integrating over the preceding record gives

$$\mathrm{Var}(\widehat{\mu}) = \mathcal{V}_{\mathrm{tr}}\mathbb{E}[1/m] + \mathcal{V}_{\mathrm{ev}}\mathbb{E}[1/(mn)]. \tag{38}$$

There is no variance of a record-dependent mean because the maintained target μ is fixed. Inverse expected counts cannot replace expected inverse counts. If the preceding record changes only the coupling within each training pair while preserving both marginal fitted-policy laws, the target remains μ, but the variance components can depend on $\mathcal{P}$. Their contribution becomes $\mathbb{E}[\mathcal{V}_{\mathrm{tr}}(\mathcal{P})/m + \mathcal{V}_{\mathrm{ev}}(\mathcal{P})/(mn)]$. Selecting the financial model or marginal training procedures can additionally change the target to $\mu(\mathcal{P})$, introducing $\mathrm{Var}\{\mu(\mathcal{P})\}$ through total variance. Equation (38) assumes fixed laws and does not supply that broader uncertainty assessment.

Unbiased variance estimation is not an exact confidence statement. For each fixed positive inner count, independent rows with positive finite variance admit the usual studentised central limit approximation as their number diverges. If the inner count is chosen from a fixed finite set using only $\mathcal{P}$, the same limiting justification applies when the selected row count diverges in probability: conditional approximation errors are uniformly small across that finite set and can then be averaged over $\mathcal{P}$. This argument gives no finite-sample coverage guarantee, no assurance for an expanding count set and no licence to collapse an interval because the observed variance is zero.

B.3 Why a cost stopping rule needs a separate argument

For a finite analytical example, let independent signs ξ_i take values -1 and 1 with equal probabilities, and let the cost of observation i be $(3+\xi_i)/2$. Stop after the first completed observation for which accumulated cost reaches two. The stopping count K is one if $\xi_1 = 1$ and two otherwise. The stopped sum has expectation zero by bounded-time optional sampling, but its sample mean satisfies

$$\mathbb{E}\left[\frac{1}{K}\sum_{i=1}^{K}\xi_i\right] = \frac{1}{2}(1) + \frac{1}{4}(0) + \frac{1}{4}(-1) = \frac{1}{4}.$$

The three terms enumerate the first positive sign, a negative followed by a positive sign, and two negative signs. The example contains no observed processor timings or market data. Its purpose is to show why dividing a stopped martingale sum by its outcome-dependent count can introduce bias. I do not infer that every CPU stopping rule is biased; rather, the fixed-count proof supplies no justification for such a rule. The example and the preceding derivations are applications of conditional expectation and optional sampling, without a claim of theoretical priority.

The accompanying `verification/allocation_probability.py` uses exact rational arithmetic on stipulated finite sampling laws to check the conditional variance identities and this stopping example. These finite calculations check an implementation of the argument; they do not verify the full measurable-space construction, certify a random-number generator or provide empirical evidence.

C Allocation uncertainty and registered contrasts

C.1 A ratio that preserves the sampling covariance

I derive a common variance coordinate to compare allocations without treating estimated components as known. Fix the financial specification and the two training procedures. For a source with m independent pairs and $n \geq 2$ evaluation paths per pair, retain the pair means W_r and put $U_r = s_r^2/n$, where s_r^2 is the within-pair sample variance of the gains Z. Then

$$v_n = \mathrm{Var}(W_r) = \mathcal{V}_{\mathrm{tr}} + \mathcal{V}_{\mathrm{ev}}/n, \qquad u_n = \mathbb{E}U_r = \mathcal{V}_{\mathrm{ev}}/n.$$

The independent units are the complete training pairs. Evaluation gains sharing a fitted pair are generally dependent after training is averaged out, so replacing the pair count by the total path count would understate uncertainty. The preceding record may select counts before fresh draws, as established in Appendix B; the new outcomes do not determine their own inclusion.

For reference count n_0 and $\kappa = n/n_0$, define

$$d_0 = v_n + (\kappa - 1)u_n = \mathcal{V}_{\mathrm{tr}} + \mathcal{V}_{\mathrm{ev}}/n_0, \quad e_0 = \kappa u_n = \mathcal{V}_{\mathrm{ev}}/n_0, \quad w = e_0/d_0.$$

The implemented reference is $n_0 = 4096$, a computational design choice. When $d_0 > 0$, $w \in [0, 1]$. If both components vanish, the ratio is undefined and every allocation has zero population variance. I estimate d_0 and e_0 by replacing v_n with S_W^2 and u_n with $\bar{U}$, retaining their signs without flooring a negative component estimate.

The identity

$$S_W^2 = \frac{m}{m-1}\left\{\frac{1}{m}\sum_r (W_r - \mu)^2 - (\bar{W} - \mu)^2\right\}$$

gives the first-order variance influence $(W_r - \mu)^2 - v_n$. Write $q_r = (W_r - \bar{W})^2$, $\bar{q} = m^{-1}\sum_r q_r$, and form the centred empirical influences

$$\widehat{\psi}_{v,r} = q_r - \bar{q}, \qquad \widehat{\psi}_{u,r} = U_r - \bar{U}, \qquad \widehat{\phi}_r = \begin{pmatrix} \widehat{\psi}_{v,r} + (\kappa - 1)\widehat{\psi}_{u,r} \\ \kappa\widehat{\psi}_{u,r} \end{pmatrix}, \qquad \widehat{\Sigma} = \frac{\sum_r \widehat{\phi}_r \widehat{\phi}_r^{\mathsf{T}}}{m(m-1)}.$$

This estimates the sampling covariance of $(\widehat{d}_0, \widehat{e}_0)$ jointly. Centring q_r by its own average matters because the unbiased sample variance has a different divisor.

Following Fieller (1954) and von Luxburg and Franz (2009), I invert the contrast rather than dividing two separate intervals:

$$(\widehat{e}_0 - w\widehat{d}_0)^2 \leq z^2\{\widehat{\Sigma}_{ee} - 2w\widehat{\Sigma}_{de} + w^2\widehat{\Sigma}_{dd}\}. \tag{39}$$

The implemented ratio routine defaults to the fixed cutoff $z = 13/4$. Under an asymptotic standard-normal contrast, a pointwise error target α would instead specify $z = \Phi^{-1}(1 - \alpha/2)$; the retained cutoff has corresponding normal tail mass $2\{1 - \Phi(13/4)\}$. This ratio acceptance set is separate from the twelve-contrast calibration reported in Table 8. Its cutoff alone supplies no simultaneous coverage claim across additional ratios or data-dependent comparisons.

The complete quadratic acceptance set is calculated on the real line before intersection with $[0, 1]$. Unbounded or disconnected sets are retained. The justification requires independent complete pairs, finite fourth moments, consistent covariance estimation and positive population variance of the true contrast. It is asymptotic as the number of pairs grows; positive empirical covariance rank is insufficient. Exact quadratic arithmetic does not certify normal approximation, finite-sample coverage or the original summary rounding errors. A weak denominator can produce a broad or disconnected unrestricted set even when the point ratio appears precise. Intersecting with the model domain restricts possible component shares; it does not supply additional observations or rescue an inaccurate normal approximation. These are applications of established ratio inference, without a claim of theoretical priority.

C.2 An endpoint comparison for the allocation rule

For feasible candidate i with m_i pairs and n_i paths per pair,

$$\operatorname{Var}(\widehat{\mu}_i) = d_0 g_i(w), \qquad g_i(w) = \frac{1-w}{m_i} + \frac{n_0 w}{m_i n_i}.$$

The common positive scale cancels when comparing candidates. I use a non-empty closed outward enclosure $[l, u] \subseteq [0, 1]$ of the retained ratio set and minimise the worst ratio to the best candidate at each w. To reduce this optimisation to a finite calculation, write $g_i = a_i + b_i w$. Every denominator is positive and

$$\frac{d}{dw}\frac{g_i(w)}{g_k(w)} = \frac{b_i a_k - a_i b_k}{(a_k + b_k w)^2}.$$

Each pairwise ratio is monotone or constant. Since $g_i / \min_k g_k = \max_k g_i/g_k$, its supremum is attained at l or u. Comparing these endpoint maxima and applying the declared tie rule determines the choice. An outward enclosure may be conservative. This argument establishes the implemented decision conditional on its set and costs; it does not establish population optimality, accurate cost forecasts or that the set covers the true ratio.

C.3 Uncertainty about a variance improvement

For an independently executed arm, $\widehat{V} = S_W^2/m$ estimates the variance of its complete-sample mean. Applying the same influence calculation gives

$$\widehat{\operatorname{Var}}(\widehat{V}) = \frac{\sum_r (q_r - \bar{q})^2}{m^3(m-1)}. \tag{40}$$

The empirical influence variance is divided by m to estimate uncertainty in S_W^2, then by m^2 for the variance of the mean. This is an asymptotic estimate, not an exact finite-sample fourth-moment formula. Independent arms permit addition of their sampling-variance estimates. Taking the square root gives the standard error of their variance difference; mean-difference standard errors instead use the sum of their estimated mean variances.

The registered intervals use the difference plus or minus 13/4 standard errors. A union bound calibrates the twelve normal-approximation contrasts without assuming independence between contrasts sharing a reference. Growing row counts, finite fourth moments and non-degenerate limiting influences are required; influential rows can make the approximation poor at the observed counts. The intervals assess differences, not variance ratios. Including zero does not establish equivalence, and no registered contrast ranks 128 against 256 paths.

References

Andersen, L. and Broadie, M. (2004). Primal-dual simulation algorithm for pricing multidimensional American options. *Management Science*, 50(9):1222–1234.

Belomestny, D., Bender, C., and Schoenmakers, J. (2009). True upper bounds for Bermudan products via non-nested Monte Carlo. *Mathematical Finance*, 19(1):53–71.

Black, F. and Scholes, M. (1973). The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3):637–654.

Cline, D. A. (2026). Efficient upper bounds for American options: Regression-based duals from backward primals. *Quantitative Finance*, 26(7):1071–1089.

Fieller, E. C. (1954). Some problems in interval estimation. *Journal of the Royal Statistical Society: Series B*, 16(2):175–185.

Glynn, P. W. and Whitt, W. (1992). The asymptotic efficiency of simulation estimators. *Operations Research*, 40(3):505–520.

Haugh, M. B. and Kogan, L. (2004). Pricing American options: A duality approach. *Operations Research*, 52(2):258–270.

Longstaff, F. A. and Schwartz, E. S. (2001). Valuing American options by simulation: A simple least-squares approach. *The Review of Financial Studies*, 14(1):113–147.

Vellekoop, M. H. and Nieuwenhuis, J. W. (2006). Efficient pricing of derivatives on assets with discrete dividends. *Applied Mathematical Finance*, 13(3):265–284.

von Luxburg, U. and Franz, V. H. (2009). A geometric approach to confidence sets for ratios: Fieller's theorem, generalizations, and bootstrap. *Statistica Sinica*, 19(3):1095–1117.